

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) Date: 20-03-26
Time: 3 Hrs Max. Marks: 300

20-03-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_GTM-11_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

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THE NARAYANA GROUP

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PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Two springs have force constants k_1 and k_2 $(k_1 > k_2)$. When they are stretched individually by the same constant force up to equilibrium, which statement is correct?

   A) No work is done by this force in case of both the springs
   B) Equal work is done by this force in case of both the springs
   C) More work is done by this force in case of second spring
   D) More work is done by this force in case of first spring

2. A small mass slides down on an inclined plane of inclination θ with the horizontal. The coefficient of friction is $\mu = \mu_0 x$ where x is the distance through which the mass slides down and μ_0 is a constant. The distance covered by the mass before it stops is:

   A) $(2/\mu_0)\tan\theta$ B) $(4/\mu_0)\tan\theta$ C) $(1/(2\mu_0))\tan\theta$ D) $(1/\mu_0)\tan\theta$

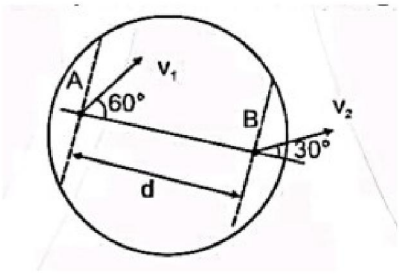
3. A sinusoidal wave with amplitude y_m is travelling with speed v on a string with linear density ρ. The angular frequency of the wave is ω. The following conclusions are drawn. Mark the one which is correct.

   A) Doubling the frequency doubles the rate at which energy is carried along the string
   B) If the amplitude were doubled, the rate at which energy is carried would be halved
   C) If the amplitude were doubled, the rate at which energy is carried would be doubled
   D) The rate at which energy is carried is directly proportional to the velocity of the wave

4. The potential energy of a particle of mass m situated in a one-dimensional potential field varies as $U(x) = U_0[1 - \cos(ax)]$ where U_0 and a are constants. The time period of small oscillations of the particle about the mean position is:

   A) $2\pi\sqrt{(m/(aU_0))}$ B) $2\pi\sqrt{(am/U_0)}$ C) $2\pi\sqrt{(m/(a^2U_0))}$ D) $2\pi\sqrt{(a^2m/U_0}$

5. Two points A and B on a disc have velocities v_1 and v_2 at some moment. Their directions make angles 60° and 30° respectively with the line of separation as shown in the figure. The angular velocity of the disc is:



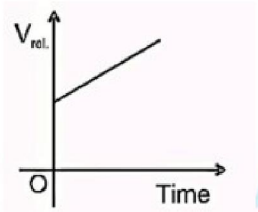
   A) $\sqrt{3}v_1/d$ B) $v_2/(\sqrt{3}d)$ C) $(v_2 - v_1)/d$ D) v_2/d

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6. For two particles A and B, given that $\vec{r}_A = 2\hat{\imath} + 3\hat{\jmath}$, $\vec{r}_B = 6\hat{\imath} + 7\hat{\jmath}$, $\vec{v}_A = 3\hat{\imath} - \hat{\jmath}$, $\vec{v}_B = x\,\hat{\imath} - 5\hat{\jmath}$. What is the value of x if they collide?

A) 1 B) −1 C) 2 D) −2

7. A particle is observed from two frames S_1 and S_2. The graph of relative velocity of S_1 with respect to S_2 is shown. Let F_1 and F_2 be the pseudo forces on the particle when seen from S_1 and S_2 respectively. Which one of the following is not possible?



A) $F_1 = 0, F_2 \neq 0$ B) $F_1 \neq 0, F_2 = 0$ C) $F_1 \neq 0, F_2 \neq 0$ D) $F_1 = 0, F_2 = 0$

8. A scientist says that the efficiency of his heat engine which operates at source temperature 127°C and sink temperature 27°C is 26%. Then:

A) It is impossible
B) It is possible but less probable
C) It is quite probable
D) Data are incomplete

9. A block of iron is kept at the bottom of a bucket full of water at 2°C. The water exerts buoyant force on the block. If the temperature of water is increased by 1°C the temperature of iron block also increases by 1°C. The buoyant force on the block by water:

A) Will increase
B) Will decrease
C) Will not change
D) May decrease or increase depending on coefficients of expansion

10. A thin uniform wire is bent to form the two equal sides AB and AC of triangle ABC, where AB = AC = 5 cm. The third side BC of length 6 cm is made from uniform wire of twice the linear mass density of the first. The distance of centre of mass from A is:

A) 34/11 cm B) 11/34 cm C) 34/9 cm D) 11/45 cm

11. A mono-atomic ideal gas is compressed from volume V to V/2 through various processes. For which of the following processes final pressure will be maximum:

A) Isobaric B) Isothermal C) Adiabatic D) PV^2 = constant

12. If R and h represent the horizontal range and maximum height respectively of an oblique projection whose start point and end point are at the same horizontal level, then $\left(\frac{R^2}{8h} + 2h\right)$ represents:

A) Maximum horizontal range
B) Maximum vertical range
C) Time of flight
D) Velocity of projectile at highest point

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13. A train A runs from east to west and another train B of the same mass runs from west to east at the same speed with respect to earth along the equator. Normal force by the track on train A is N_1 and that on train B is N_2 :

A) $N_1 > N_2$

B) $N_1 < N_2$

C) $N_1 = N_2$

D) Information is insufficient to find relation

14. The two plates of a parallel plate capacitor with no dielectric are connected to a battery with constant emf. Now a dielectric of dielectric constant K is inserted to fill the entire space between the plates without removing the voltage source. Find the **INCORRECT** option.

A) The energy stored in the capacitor will become K times.

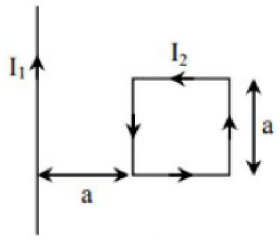
B) The electric field between the plates will become 1/K times.

C) The force of attraction between plates will become K^2 times.

D) Charge on capacitor will become K times.

15. A rigid square loop of side 'a' and carrying current I_2 is lying on a horizontal surface near a long wire carrying current I_1 in the same plane as shown in figure.

The net force on the loop due to the wire will be



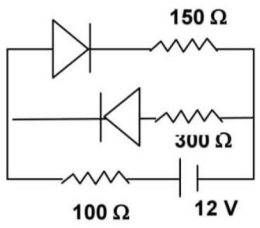
A) Repulsive and equal to $\frac{\mu_0 I_1 I_2}{2\pi}$

B) Attractive and equal to $\frac{\mu_0 I_1 I_2}{3\pi}$

C) Zero

D) Repulsive and equal to $\frac{\mu_0 I_1 I_2}{4\pi}$

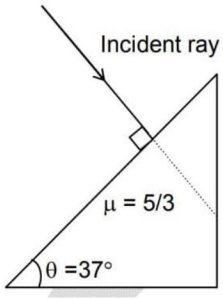
16. What is the current in the circuit?



A) 60 Ma B) 30 mA C) 15 mA D) 48 mA

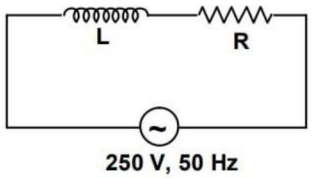
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17. A light ray is incident perpendicularly on a face of a right angle prism as shown in the figure. The total angle of deviation of the ray is (sin 37° = 3/5).



A) 74° B) 106° C) 127° D) 53°

18. A series L–R circuit is connected across a source of 250 V, 50 Hz supply. The power factor of the circuit is 0.8 and power of 400 W is dissipated. The value of inductive reactance and resistance are:



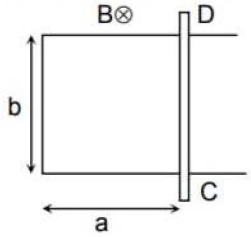
A) $X_L = 75\ \Omega,\ R = 100\ \Omega$ B) $X_L = 1000\ \Omega,\ R = 75\ \Omega$

C) $X_L = 70\ \Omega,\ R = 1000\ \Omega$ D) $X_L = 50\ \Omega,\ R = 1000\ \Omega$

19. Light of energy 2 eV and intensity 8 W/m² falls on a metal surface of area $10 cm^2$ and work function 1.5 eV. Only 20% of incident photons produce photoelectrons. The number of emitted photoelectrons per second will be:

A) 4×10^{17} B) 5×10^{17} C) 5×10^{15} D) 5×10^{18}

20. A U-shaped conducting frame is fixed in space. A conducting rod CD lies at rest on the smooth frame. The frame is in a uniform magnetic field B_0 perpendicular to the plane of the frame. At time t = 0 the magnetic field changes with time as



$B = B_0 / (1 + kt)$ For no current to be induced in the frame, find the speed with which the rod should be pulled starting from time t = 0.

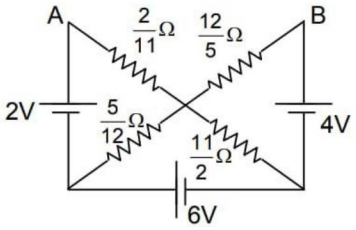
A) ak B) bz C) a(1 + kt) D) b(1 + et)

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. In the circuit shown, the value of $V_A - V_B =$ _____ V.



22. A parallel plate capacitor has square plates of side length L kept at a separation d. The space between them is filled with a dielectric constant which changes as $k = B/x^2$, where B is a positive constant and x is the distance measured from left plate towards right plate. Equivalent capacitance is:

$C = (NBL^2\varepsilon_0)/d^3$ Find the value of N.

23. The electric field of a plane electromagnetic wave is given by (speed of light $c = 3\times10^8 m/s$) $E = 100\sin[2\pi\times2\times10^{15}\left(t-\frac{X}{C}\right)]V/m$. If this wave falls normally on a perfectly reflecting surface having an area of 10 cm². The radiation pressure exerted by the electromagnetic wave on the surface during a 10 minute of exposure is $\frac{X}{10^{10}}N/m^2$ Fin the value of x. (permittivity of free space $\varepsilon_0 = 8.85\times10^{-12}C^2N^{-1}m^{-2}$)

24. In Young's double slit experiment, a monochromatic light of wavelength λ is used. The intensity of light at a point on screen where the path difference is λ is I_0 units. The intensity of light on the screen where the path difference $\lambda/4$ is given by I_0/n units. Find the value of n.

25. In the line spectra of hydrogen atom, difference between the largest and the shortest wavelengths of the Lyman series is $301\ \mathring{A}$. The corresponding difference for the Paschen series (in $\mathring{A}$) is

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Match List – I with List – II

	List – I		**List – II**
(a)	Lassaigne's test	(i)	Carbon
(b)	Cu(II) oxide	(ii)	Sulphur
(c)	Silver nitrate	(iii)	N,S,P and halogen
(d)	The sodium fusion extract gives black precipitate with acetic acid and lead acetate	(iv)	Halogen specifically

The correct match is

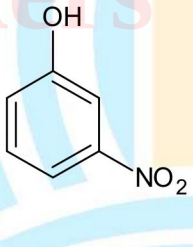
A) a – iii, b – i, c – iv, d – ii B) a – i, b – iv, c – iii, d – ii

C) a – i, b – ii, c – iv, d – iii D) a – iii, b – i, c – ii, d – iv

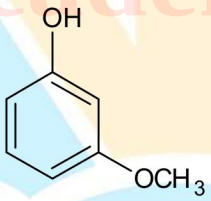
27. Among the following in decreasing acidic strength



I



II



III



IV

A) I>II>III>IV B) II>I>III>IV C) IV>III>I>II D) IV>III>II>I

28. 20 ml of 0.1 M NH_4OH is mixed with 40 ml of 0.05 M HCl. The pH of the mixture is

[Given K_b $(NH_4OH)=1x10^{-5}$, log2=0.30, log3=0.48, log5=0.69, log7=0.84, log11=1.04)

A) 3.2 B) 4.2 C) 5.2 D) 6.2

29. In chromatography technique the purification of compound is independent of

A) Solubility of the compound

B) Length of the column or TLC plate

C) Mobility or flow of solvent system

D) Physical state of the pure compound

For More Join: @IITJEE_Advanced

30. **Assertion:** When aniline is subjected to nitration by conc. HNO_3 & H_2SO_4 meta nitro aniline is formed in considerable amount.

Reason : NH_2 is o/p directing but ring deactivating group.

A) Assertion is True, Reason is true: Reason is the correct explanation for Assertion.

B) Assertion is true Reason is true : Reason is not the correct explanation for Assertion.

C) Assertion is True, Reason is False.

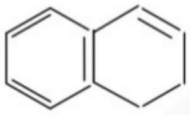
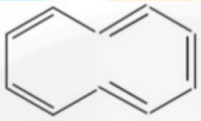
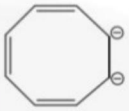
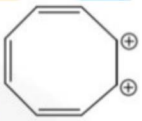
D) Assertion is False, Reason is true.

31. For the process : $H_2O(l)(1\ atm, 373.15K) \rightarrow H_2O(g)(1atm, 373.15K)$, the correct set of thermodynamic parameter is

A) $\Delta G = 0; \Delta S > 0$ B) $\Delta G = 0; \Delta S < 0$

C) $\Delta G > 0; \Delta S = 0$ D) $\Delta G < 0; \Delta S > 0$

32. Which of the following is/are aromatic in character?

(I)  (II)  (III)  (IV) 

A) II only B) I, III and IV

C) II, III and IV D) I, II, III and IV

33. Which of the following is incorrect?

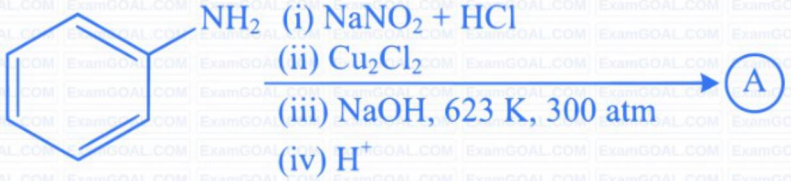
A) PCl_5 exists as $[PCl_4^+][PCl_6^-]$ in solid state

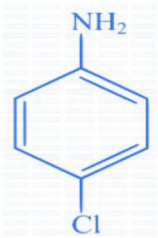
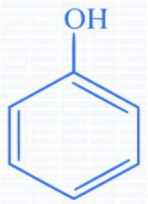
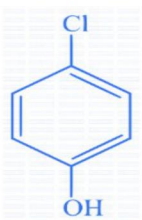
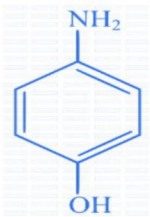
B) PBr_5 exists as $[PBr_4^+][PBr_6^-]$ in solid state

C) Bond order of O_2 is 2

D) All the P-Cl bond lengths in PCl_5 are not same

34. Identify the product A in the following reaction.



A)  B)  c)  D) 

For More Join: @IITJEE_Advanced

35. Given below are two statements:

Statement I: According to Bohr's model of an atom, qualitatively the magnitude of velocity of electron increases with decrease in positive charge on the nucleus as there is no strong hold on the electron by the nucleus.

Statement II: Orbital angular momentum of 3s orbital is zero

A) Statement I is true but Statement II is false

B) Statement I is false but Statement II is true

C) Both Statement I and Statement II are false

D) Both Statement I and Statement II are true

36. Which one of the following complexes is violet in colour?

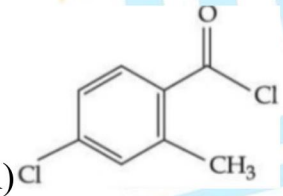
A) $[Fe(SCN)_6]^{4-}$ B) $[Fe(CN)_6]^{4-}$

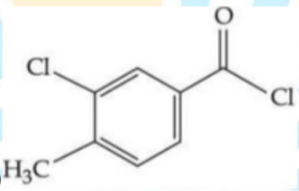
C) $Fe_4[Fe(CN)_6]_3.H_2O$ D) $[Fe(CN)_5NOS]^{4-}$

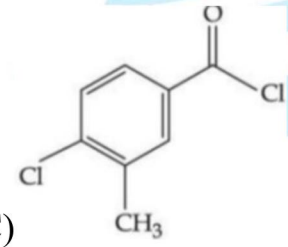
37.

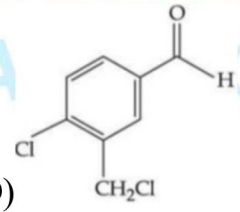


Consider the above reaction, the compound 'A' is:

A) 

B) 

C) 

D) 

38. Select the incorrect statement(s) among the followings

(1) The unit of rate constant for second order reaction is $mol^{-1}Ls^{-1}$

(2) Inversion of cane sugar is a pseudo first order reaction

(3) Order of a reaction can only be a whole number

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(4) The exponents of concentration terms in rate law expressions are always equal to the stoichiometric coefficients of the reactions

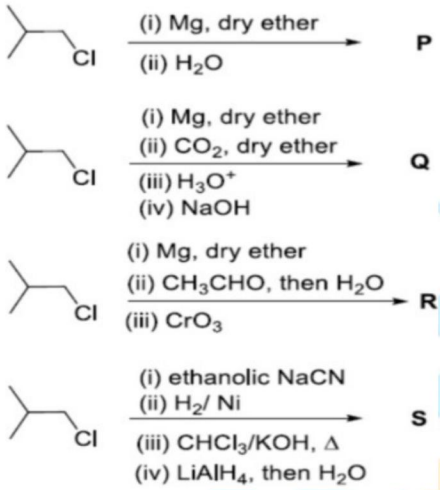
A) (1) and (2) only

B) (2) and (4) only

C) (3) and (4) only

D) (1) and (3) only

39. In the following reactions, P,Q,R and S are the major products.



The correct statement about P,Q,R and S is

A) P is a primary alcohol with four carbons.

B) Q undergoes Kolbe's electrolysis to give an eight-carbon product.

C) R has six carbons and it undergoes Cannizzaro reaction

D) S is a primary amine with six carbons.

40. Given below are two statements:

Statement I : The number of paramagnetic species among $[CoF_6]^{3-}$, $[TiF_6]^{3-}$, V_2O_5 and $[Fe(CN)_6]^{3-}$ is 3 .

Statement II :

$K_4[Fe(CN)_6] < K_3[Fe(CN)_6] < [Fe(H_2O)_6]SO_4 \cdot H_2O < [Fe(H_2O)_6]Cl_3$ is the correct order in terms of number of unpaired electron(s) present in the complexes.

In the light of the above statements, choose the correct answer from the options given below

A) Statement I is true but Statement II is false

B) Both Statement I and II are false

C) Statement I is false but Statement II is true

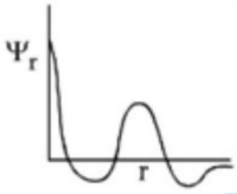
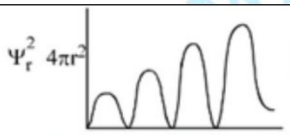
D) Both Statement I and II are true

For More Join: @IITJEE_Advanced

41. During the borax bead test with $CuSO_4$ a blue green colour of the bead was observed in oxidizing flame due to the formation of

A) Cu_3B_2 B) Cu C) $Cu(BO_2)_2$ D) CuO

42. Column-I & Column-II contain data on Schrondinger Wave-Mechanical model, where symbols have their usual meanings. Match the columns.

	Column – I		Column – II
1)		P)	4s
2)		Q)	$5p_x$
3)	$\psi(\theta,\phi) = K$ (independent of θ & ϕ)	R)	3s
4)	At least one angular node is present	S)	$6d_{xy}$

A) 1 – P, 2 – PQS, 3 – PR, 4 – QS B) 1 – PQS, 2 – P, 3 – QS, 4 – PR

C) 1 – PQS, 2 – QS, 3 – P, 4 – PR D) 1 – QS, 2 – PQS, 3 – PR, 4 – P

43. Calculate the value of $\Delta H^\circ / kJ$ for the following reaction $2C(s) + H_2(g) \rightarrow C_2H_2(g)$ by using the listed thermochemical equations:

$2C_2H_2(g) + 5O_2(g) \rightarrow 4CO_2(g) + 2H_2O(l)\ \Delta H^\circ / kJ = -2600kJ$

$C(s) + O_2(g) \rightarrow CO_2(g)\ \Delta H^\circ / kJ = -390kJ$

$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)\ \Delta H^\circ / kJ = -572\,kJ$

A) +184 B) +214 C) +202 D) +234

44. The lanthanoid that does not shows +4 oxidation state is

A) Tb B) Dy C) Eu D) Ce

45. What are the functional groups present in the structure of maltose?

A) Two acetals B) One ketal and one hemiketal

C) One acetal and one ketal D) One acetal and one hemiacetal

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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. How many of the following ethers cannot be prepared by Williamson's synthesis are?

$CH_3OCH_2CH_3$, $C_6H_5OCH_3$, $(C_6H_5)_2O$

$(CH_3)_3COCH_3$, $C_6H_5CH_2OCH_3$, $(C_2H_5)_2O$,

$C_6H_5CH_2OC_6H_5$, $CH_3CH_2OCH_2CH_3$, $(CH_3)_3C-O-C(CH_3)_3$

47. The number of Cl = O bonds in perchloric acid is, "_____"

48. The elevation of boiling point of 0.10 m aqueous $CrCl_3.xNH_3$ solution is two times that of 0.05 m aqueous $CaCl_2$ solution. The value of x is _____.

[Assume 100% ionization of the complex and $CaCl_2$, coordination number of Cr as 6, and that all NH_3 molecules are present inside the coordination sphere].

49. If x and y are total number of electrons which are present in non-axial and axial set of orbitals respectively in Ni cation of $[Ni(CO)_4]$, then calculate value of $\left(\frac{x}{y}\right)\times 2$ is

50. $Pt(s)\,|\,H_2(g)(1bar)\,|\,H^+(aq)(1M)\,||\,M^{3+}(aq), M^{1+}(aq)\,|\,Pt(s)$ The E_{cell} for the given cell is 0.1115V at 298K when $\frac{[M^+(aq)]}{[M^{3+}(aq)]}=10^a$ The value of a is _____.

Given $E°M^{3+}/M^+ = 0.2V$, $\frac{2.303RT}{F}=0.059V$

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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If the foot of the perpendicular form the point $A(-1,4,3)$ on the plane $P: 2x+my+nz=4$, is $\left(-2,\frac{7}{2},\frac{3}{2}\right)$, then the distance of the point A form the plane P, measured parallel to a line with direction ratios $3,-1,-4$, is equal to

A) 1 B) $\sqrt{26}$ C) $2\sqrt{2}$ D) $\sqrt{14}$

52. Let $f: R \to R$ is defined by $f(x)=\frac{e^x.(x-3)}{\left(x^2+x+\frac{3}{2}\right)}$

A) f(x) is increasing in $(0,\infty)$

B) f(x) is decreasing in $(0,\infty)$

C) f(x) is increasing in $(-\infty,0)$

D) f(x) is increasing in $(-\infty,\infty)$

53. The area of the region $S=\{(x,y):(x-2024)[x]\le y\le 4\sqrt{x};0\le x\le 2\}$; (where [x] represents greatest integer function) is

A) $\frac{16\sqrt{2}}{3}$ B) $\frac{16\sqrt{2}}{3}+2022$

C) $\frac{16\sqrt{2}}{3}+\frac{4045}{2}$ D) $\frac{8\sqrt{2}}{3}+\frac{4045}{2}$

54. The coefficient of x^4 in the expansion of $(1+x+x^2+x^3)^{11}$ is (where $\binom{n}{r}={}^nC_r$)

A) $\binom{11}{4}$ B) $\binom{11}{4}+\binom{11}{2}$

C) $\binom{11}{4}+\binom{11}{2}+\binom{11}{4}.\binom{11}{2}$ D) $\binom{11}{4}+\binom{11}{2}+\binom{11}{1}\binom{11}{2}$

55. If the equation of directrix of a rectangular hyperbola is x + y = 1; whose centre is (0, 5); is $ax^2+by^2+4xy+cx+dy+e=0$ then $c+d+2e$ equals

A) -58 B) $-46+4\sqrt{2}$ C) $-46-4\sqrt{2}$ D) -84

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56. Let α and β be complex numbers with imaginary part 1 such that for all complex numbers $z = x + iy$ satisfying $\text{Arg}\left(\frac{z-\alpha}{z-\beta}\right) = \frac{\pi}{4}$, and the point (x, y) lies on the circle $x^2 + y^2 - 8x - 6y + 17 = 0$, then $|\alpha\beta|$ is

A) $\sqrt{175}$ B) $\sqrt{185}$ C) $\sqrt{195}$ D) 15

57. Let $y = y(x), y > 0$ be a solution curve of the differential equation $(1+x^2)dy = y(x-y)dx$. If $y(0) = 1$ and $y(2\sqrt{2}) = \beta$, then

A) $e^{3\beta^{-1}} = e(3+2\sqrt{2})$ B) $e^{\beta^{-1}} = e^{-2}(5+\sqrt{2})$

C) $e^{\beta^{-1}} = e^{-2}(3+2\sqrt{2})$ D) $e^{3\beta^{-1}} = e(5+\sqrt{2})$

58. Domain of derivative of the function $f(x) = \begin{cases} \frac{x}{1+|x|}, |x| \geq 1 \\ \frac{x}{1-|x|}, |x| < 1 \end{cases}$ is

A) $R - \{-1, 0, 1\}$ B) $R - \{-1, 1\}$ C) $R - \{0\}$ D) R

59. $\lim\limits_{m\to\infty} \int\limits_{-\infty}^{\infty} \frac{dx}{1+x^2+x^4+\ldots x^{2m}} (m = 1, 2, \ldots)$

A) 4 B) $\frac{4}{3}$ C) $\frac{3}{4}$ D) 0

60. If $\int\limits_0^{\pi} \left(\frac{x}{1+\sin x}\right)^2 dx = A$, then the value of $\int\limits_0^{\pi} \frac{2x^2.\cos^2(x/2)}{(1+\sin x)^2} dx$ is equal to

A) $A + 2\pi - \pi^2$ B) $A - 2\pi - \pi^2$ C) $2\pi - A - \pi^2$ D) $A + 2\pi + \pi^2$

61. If $\int \frac{\sec\theta d\theta}{\tan^3\theta(1+\sec^2\theta)^2} = \frac{1}{4\sec^3\theta(1-\sec^4\theta)} - \frac{3}{4}F(\theta)$ and $F\left(\frac{\pi}{3}\right) = -\frac{1}{4}\ln\alpha - \frac{1}{2}\tan^{-1}\beta + \frac{1}{\gamma}$, then the value of $\alpha + \beta + \gamma =$

A) 14 B) 29 C) 19 D) 0

62. Let D be the domain of the function $f(x) = \sin^{-1}\left(\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\right)$

If the range of the function $g: D \to \mathbb{R}$ defined by $g(x) = x - [x]$, ([x] is the greatest integer function), is $[\alpha, \beta)$, then $[\log_3 \alpha]$ is

A) -12 B) -8 C) -4 D) -2

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63. Sum of series of $\cot^{-1}\frac{5}{\sqrt{3}}+\cot^{-1}\frac{9}{\sqrt{3}}+\cot^{-1}\frac{15}{\sqrt{3}}+\cot^{-1}\frac{23}{\sqrt{3}}+\ldots\infty$ is equal to

A) $\frac{\pi}{4}$ B) $\frac{\pi}{3}$ C) $\frac{\pi}{6}$ D) $\frac{\pi}{12}$

64. The number of ways of choosing 10 objects out of 31 objects of which 10 are identical and the remaining 21 are distinct, is:

A) 2^{20} B) $2^{20}+1$ C) 2^{21} D) $2^{20}-1$

65. Let $A=\begin{bmatrix}1 & \frac{1}{51}\\ 0 & 1\end{bmatrix}$. If $B=\begin{bmatrix}1 & 2\\ -1 & -1\end{bmatrix}A\begin{bmatrix}-1 & -2\\ 1 & 1\end{bmatrix}$, then the sum of all the elements of the matrix $\sum_{n=1}^{50}B^n$ is equal to

A) 125 B) 100 C) 75 D) 50

66. Let A = {1, 2, 3, 4, 5, 6, 7}. Then the relation $R=\{(x,y)\in A\times A: x+y=7\}$ is

A) an equivalence relation

B) reflexive but neither symmetric nor transitive

C) transitive but neither symmetric nor reflexive

D) symmetric but neither reflexive nor transitive

67. Let the mean and variance of 12 observations be $\frac{9}{2}$ and 4 respectively. Later on, it was observed that two observations were considered as 9 and 10 instead of 7 and 14 respectively. If the correct variance is $\frac{m}{n}$, where m and n are coprime, then m + n is equal to

A) 316 B) 317 C) 314 D) 315

68. Let A, B, C be three events. Given below are two statements

$S_1: P\left(\frac{A}{C}\right)+P\left(\frac{B}{C}\right)$ can be greater than 1

$S_2: P\left(\frac{A\cup B}{C}\right)$ can be greater than 1,

then

A) only S_1 is true B) only S_2 is true

C) both S_1 and S_2 are true D) neither S_1 nor S_2 is true

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69. The image of the point (3, 5) in the line $x - y + 1 = 0$, lies on

A) $(x-2)^2 + (y-4)^2 = 4$ B) $(x-2)^2 + (y-2)^2 = 12$

C) $(x-4)^2 + (y-4)^2 = 8$ D) $(x-4)^2 + (y+2)^2 = 16$

70. Let $f(\theta) = \frac{1}{1+(\cot\theta)^x}$ and $S = \sum_{\theta=1^\circ}^{89^\circ} f(\theta)$, then [S] is. (where [.] denotes greatest integer function)

A) 44 B) 45 C) 46 D) 43

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. Let L_1 be a line in xy-plane with x and y intercepts 8 and 6 respectively and L_2 be a line in yz plane with y and z intercepts -8 and -5 respectively. If d is the shortest distance between L_1 and L_2 then $\frac{1649}{280}d^2$ is

72. Let $S_1 : x^2 + y^2 + 4x - 6y - 12 = 0$, and $S_2 : x^2 + y^2 - 3x - 2y + 1 = 0$ be two distinct circles with centres C_1 and C_2 respectively. If A and B are two points which lie on their common chord as well as common tangents and $C_1A^2 - C_2B^2 = \frac{a}{b}$, where $a, b \in N$ and a, b are co-prime. Then a + b is _____.

73. Let *m* and *n* be the numbers of real roots of the equations $x^2 - 12x + [x] + 31 = 0$ and $x^2 - 5|x+2| - 4 = 0$ respectively, where [x] denotes the greatest integer $\le x$. Then $m^2 + mn + n^2$ is equal to _____.

74. The value of $\sum_{m=0}^{\infty}\sum_{n=0}^{\infty}\frac{m+n+mn}{2^m(2^m+2^n)}$ is equal to

75. Let $\vec{a} = 2\hat{i} - 5\hat{j} + 5\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + 3\hat{k}$. If $\vec{c}$ is a vector such that $2(\vec{a}\times\vec{c}) + 3(\vec{b}\times\vec{c}) = \vec{0}$ and $(\vec{a} - \vec{b}).\vec{c} = -97$, then $|\vec{c}\times\hat{k}|^2$ is equal to

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